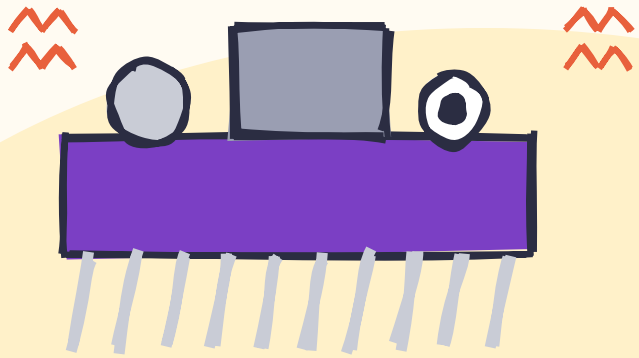
Little Explorers · Class 11

Vibrating Brush Robot

A tiny robot with no wheels that shakes to move!

Vibration motor | Balance | Build, test, improve



Did you know?
Your phone has a tiny vibration motor inside. That is what buzzes when it rings on silent!

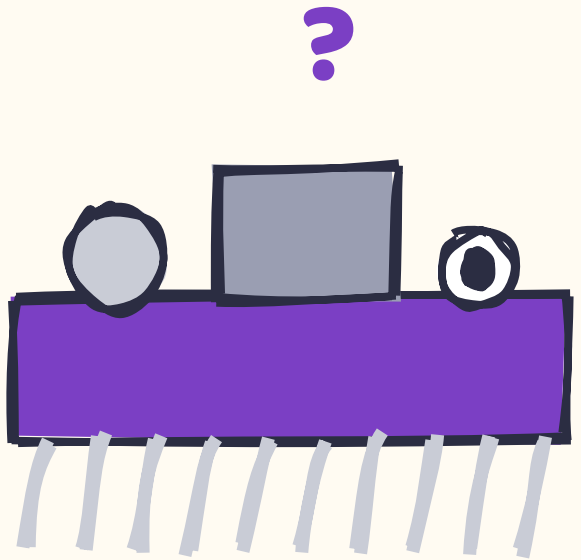
LEARN

Let's wonder...

Can a robot walk without wheels or legs?

A robot puzzle

Most robots have wheels or legs. Today's robot has **none**! It only has a brush and a tiny motor. How will it move?



It can't!

Someone pushes it?

It shakes!

Today: we build a **brush bot** and race it in an arena!

Learning objectives

Today I can...

1	**Vibration:** say that a **shaking** motor can make a robot move	**Proof:** show the motor buzzing and the robot moving
2	**Build:** build a **brush robot** with a motor and a battery	**Proof:** my brush bot moves on the table
3	**Improve:** change the **weight** or shape to make it move differently	**Proof:** I changed one thing and saw a difference

Objective 1 · Vibration · video

Video time!



Bristlebot: Build Your Own

Keen Kids · about 5 minutes · youtu.be/y5U5pOZWlkw

Watch for...

1. What makes the bristlebot **move**?
2. What do the bristles (brush hairs) do?
3. Which way does it go?

Scan to watch

After the video: Put your hands on the table and shake them very fast. Do they slide? That is how a brush bot moves!

Objective 1 · Vibration

Shaking makes it move

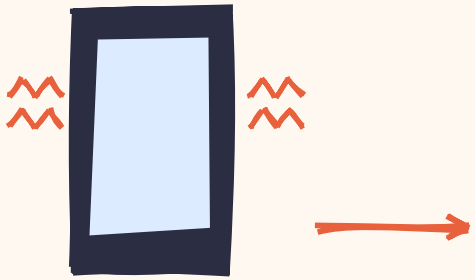
The idea

The tiny motor has a **wonky weight** on it. When it spins, the whole robot **shakes** very fast (vibrates).

The brush hairs are **bent**, so every shake pushes the robot a tiny step forward. Lots of tiny steps = it zooms!

New word
Vibrate means to shake back and forth very, very fast.

Example: a phone on silent



When a phone rings on silent, it **buzzes** and sometimes slides across the table all by itself. Same idea: a tiny motor shaking!

LEARN

Objective 3 · Improve

Build, test, improve

The idea

Engineers almost never get it right the first time. They **build**, **test**, then **improve** one thing.

Try moving the motor to the **back** or the **side**, or add a small **weight** like a coin. Does your robot go straighter? Does it spin?

Engineer trick
Change ONE thing at a time, so you know what made the difference.

Example: a paper aeroplane



Your first paper plane might nose-dive. You fold the wings a little differently and throw again. It flies further! That is **improving**.

Real world

Vibration is everywhere



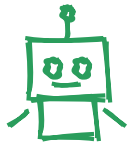
Phones
Buzz when a message arrives



Game controllers
Rumble when you crash in a game



Smartwatches
Tap your wrist to wake you up



Tiny robots
Swarms of shaking robots help scientists study ants and bees

? Can you think of a toy that shakes or buzzes?

Swarms
Scientists build hundreds of tiny robots that work together like ants!

Part 2 · Make it

Now let's put it to work!

You know the ideas. Time to play, build and test. Use these slides while you work:

1. Our brush bot kit
2. Build your brush bot
3. Brush bot dance
4. The great brush bot race
5. What changes the way it moves?
6. Brush bot challenges

Build it

Our brush bot kit



Brush head
The robot's body and feet



Vibration motor
Makes the shaking



Coin battery
Gives the power



Googly eyes
So it looks friendly!



Sticky tape
Holds it together



Plate arena
Where we race

Coin batteries are small but **not for mouths**. Always give them back to the teacher after class.

Safety
Never put a coin battery in your mouth, nose or ears.

MAKE

Objective 2 · Build step 1 of 2

Build your brush bot

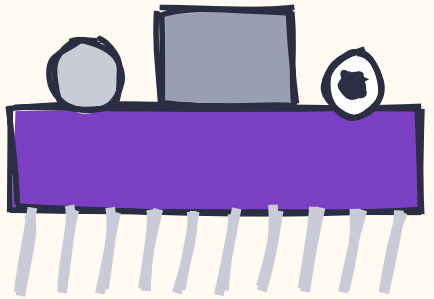
a Stick the **double-sided tape** on top of the brush head.

b Press the **motor** on top, near the front.

c Slide the **coin battery** under one wire of the motor.

d Touch the other wire to the battery: **bzzzz!**

e Add googly eyes and give it a name!



Check: The motor buzzes when both wires touch the battery.

Coin batteries must go back to the teacher after class.

Class game

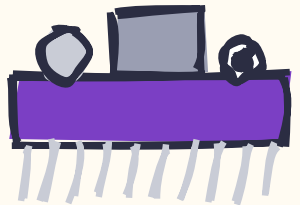
Brush bot dance

How to play

1. Children are brush bots: stand with feet together.
2. Teacher says **"Switch on!"**: shake your whole body.
3. Shuffle in tiny steps without lifting your feet.
4. **"Switch off!"**: freeze like a statue!
5. Who travelled furthest?

Then discuss

Was it easy to go **straight**? Did anyone spin around?



Big idea

Tiny shakes, added up, become movement.

Objective 2 · Build · race time

The great brush bot race

Try	Time (seconds)	My robot went…
Race 1	_____	straight / spin / stuck
Race 2	_____	straight / spin / stuck
Race 3	_____	straight / spin / stuck

How to race

Put your bot in the middle of the plate. Who reaches the **edge** first?

Fair race
Start together when the teacher says GO!

Improve it!

What changes the way it moves?

Change	I think…	What happened
Motor at the front	straight / spin?	_____
Motor at the side	straight / spin?	_____
Add a coin on top	faster / slower?	_____
Short bristles vs long	faster / slower?	_____

Talk about it

Which change made your robot go **straightest**? Which made it **spin**?

Draw it

Draw your best robot design so you remember it.

Level up!

Brush bot challenges

Choose your challenge



Spinner: make your robot spin in circles on purpose.



Team bot: join two brush heads together. What happens?



Maze: build a maze with books. Can your robot get through?



Art bot: add a felt-tip pen. Can it draw a pattern?

Think like an engineer

Why do you think brush bots are hard to steer?

Boss level

Make your robot go straight across the whole arena!

Objective check

Can I really do it? Prove it.

01	What makes the brush bot move? Use the word **vibrate**.	○ ○ ○
02	Show your brush bot moving on the table.	○ ○ ○
03	Tell us one thing you changed and what happened.	○ ○ ○

Rate yourself

Colour one circle each. Red: not yet. Amber: with help. Green: I can explain it to a friend.

Teacher: Gallery: line up all the robots and vote for the fastest, the straightest and the best named.

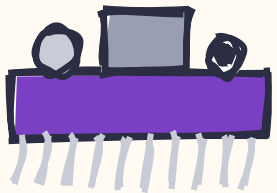
Show what you know

What did we learn?

- [x] A vibration motor shakes very fast
- [x] Bent bristles turn shakes into steps
- [x] Changing weight or motor position changes the movement
- [x] Engineers build, test and improve

At home

Find something at home that vibrates. Tell your family how it works!



Next class: send secret messages with Morse Code!

Exit ticket: why did your robot spin (or go straight)?